Angle Modulation

The Angle of the Carrier

- Let $\theta_i(t)$ represent the instantaneous angle of the carrier.
- We express an angle modulated signal by:

$$s(t) = A_c \cos\left(\theta_i(t)\right)$$

where A_c is the carrier amplitude.

Instantaneous Frequency

- One cycle occurs when $\theta_i(t)$ changes by 2π radians.

$$f_{\Delta t} = \frac{\theta_i(t + \Delta t) - \theta_i(t)}{2\pi \Delta t}$$

- Therefore, the instantaneous frequency is found in the limit as $\Delta t \to 0$:

$$f_i(t) = \frac{1}{2\pi} \frac{d\theta_i(t)}{dt}$$

Phase Modulation (PM)

- There are two angle modulation techniques:
  - Phase Modulation (PM)
  - Frequency Modulation (FM)
- In PM, the phase of the carrier is a linear function of the message signal $m(t)$.

$$s_{PM}(t) = A_c \cos\left(2\pi f_c t + k_p m(t) + \phi_c\right)$$

where k_p is the phase sensitivity.

Assuming $\phi_c = 0$:

$$\theta_i(t) = 2\pi f_c t + k_p m(t)$$

Frequency Modulation (FM)

- For FM, the instantaneous frequency is:

$$f_i(t) = f_c + k_f m(t)$$

where k_f is the frequency sensitivity.

$$\theta_i(t) = 2\pi \int_{-\infty}^{t} f_i(\tau) d\tau$$

$$\theta_i(t) = 2\pi f_c t + 2\pi k_f \int_{-\infty}^{t} m(\tau) d\tau$$

$$s_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int_{-\infty}^{t} m(\tau) d\tau\right]$$

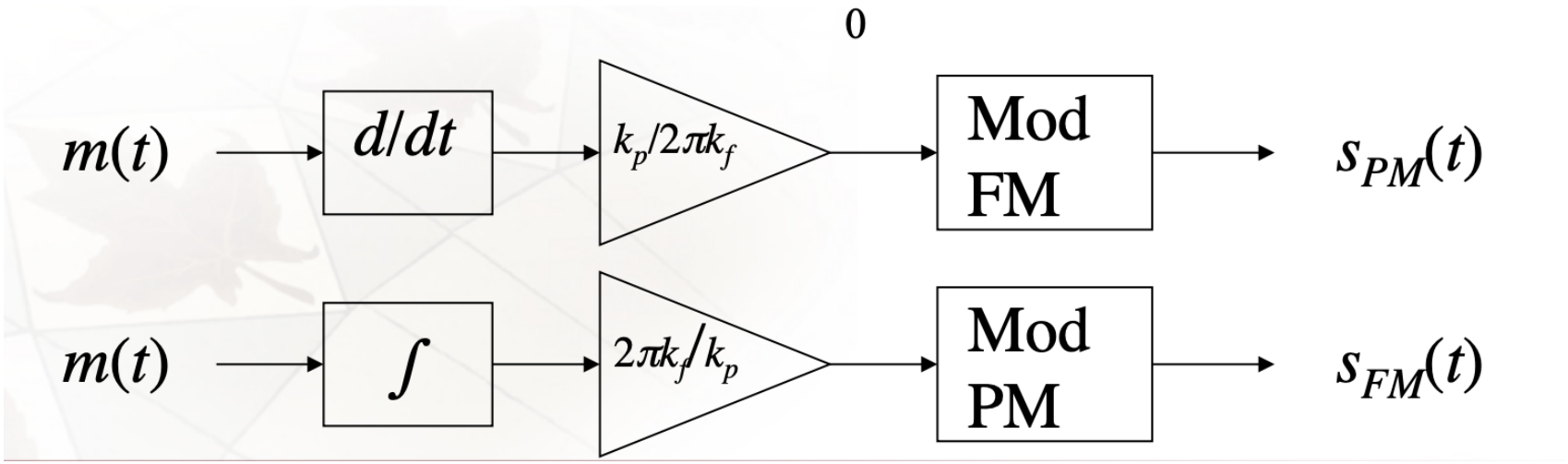
Instantaneous Frequency of PM / Instantaneous Phase of FM

From $s_{PM}(t)$:

$$f_i(t)_{PM} = f_c + \frac{k_p}{2\pi}\frac{dm(t)}{dt}$$

From $s_{FM}(t)$:

$$\phi_i(t)_{FM} = 2\pi k_f \int_0^t m(\tau)d\tau$$


0
m(t)
d/dt
k_p/2πk_f
Mod
FM
s_PM(t)
m(t)
∫
2πk_f/k_p
Mod
PM
s_FM(t)

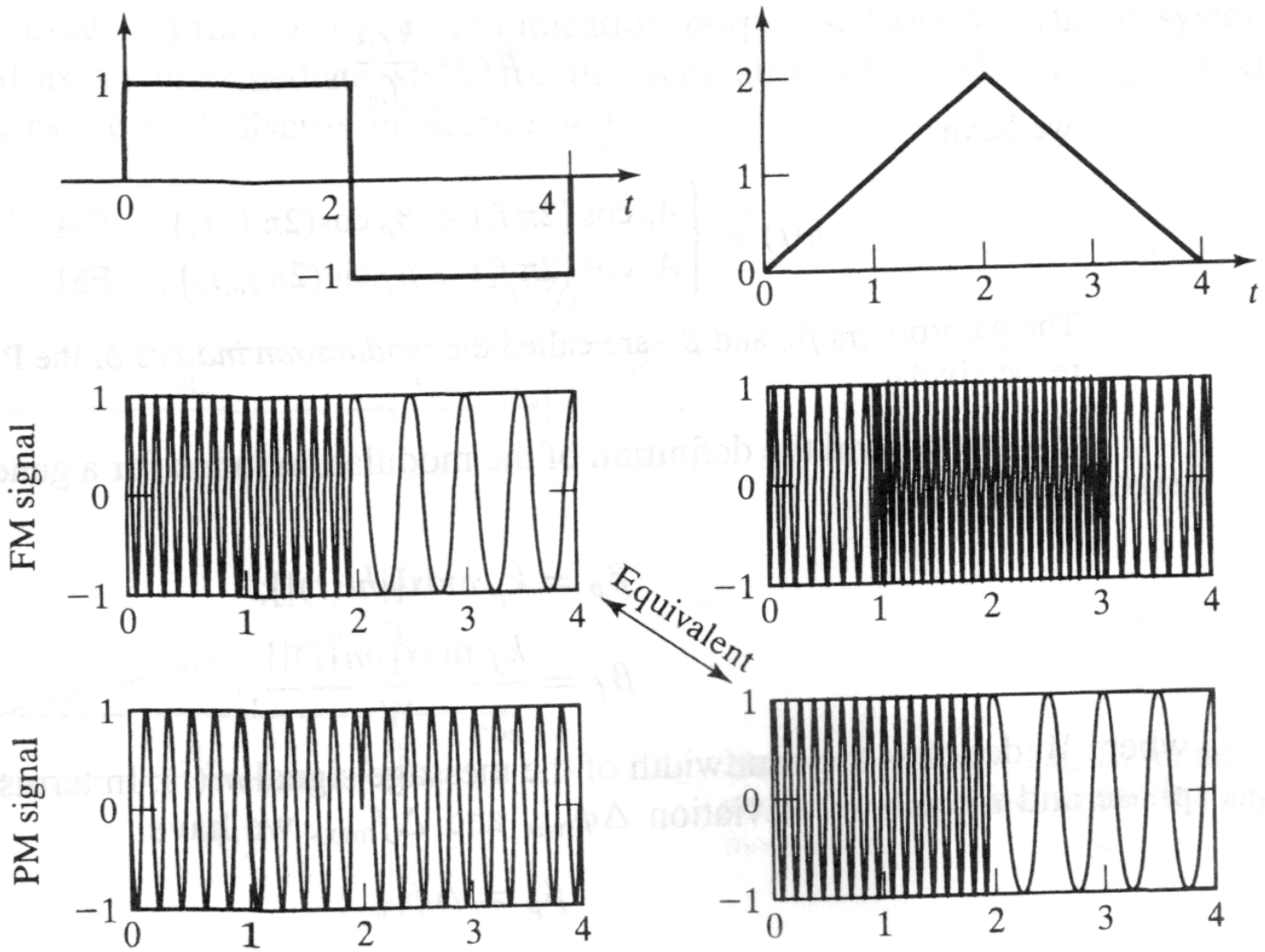


Figure 4.2 Frequency and phase modulation of square and sawtooth waves.

Example

Find $s_{FM}(t)$ and $s_{PM}(t)$ if

$$m(t) = A\cos(2\pi f_m t)$$

Solution

$$s_{PM}(t) = A_c \cos\left[2\pi f_c t + Ak_p \cos(2\pi f_m t)\right]$$

$$\int A\cos(2\pi f_m t)dt = \frac{A}{2\pi f_m}\sin(2\pi f_m t)$$

$$s_{FM}(t) = A_c \cos\left[2\pi f_c t + \frac{Ak_f}{f_m}\sin(2\pi f_m t)\right]$$

Characteristics of Angle Modulated Signals

	PM Signal	FM Signal
Instantaneous phase $\phi_i(t)$	$k_p m(t)$	$2\pi k_f \int_0^t m(\tau)d\tau$
Instantaneous frequency	$f_c + \frac{k_p}{2\pi}\frac{dm(t)}{dt}$	$f_c + k_f m(t)$
Maximum phase deviation	$k_p \lvert m(t)\rvert_{max}$	$2\pi k_f \lvert x(t)\rvert_{max}$
Maximum frequency deviation	$\frac{k_p}{2\pi}\lvert x(t)\rvert_{max}$	$k_f \lvert m(t)\rvert_{max}$
Power	$\frac{A_c^2}{2}$	$\frac{A_c^2}{2}$

Modulation Index

Assume:

$$m(t) = A_m \cos(2\pi f_m t)$$

Then FM signal:

$$s_{FM}(t) = A_c \cos\left(2\pi f_c t + \frac{A_m k_f}{f_m} \sin(2\pi f_m t)\right)$$

$$\beta_F = \frac{k_f A_m}{f_m} = \frac{\Delta f_{max}}{f_m}$$

FM Modulation Index

- For any $m(t)$ which has bandwidth B_m, we define the modulation index as:

$$\beta_F = \frac{k_f |m(t)|_{\max}}{B_m} = \frac{\Delta f_{\max}}{B_m}$$

Example

- The signal $m(t) = 5\,\mathrm{sinc}^2(10t)$.

Find the modulation index for FM modulation with $k_f = 20$ Hz/V.

Solution

- $B_m = 10$ Hz, therefore

$$\beta_F = \frac{20 \times 5}{10} = 10.$$

Narrowband FM

Consider an FM signal:

$$s_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int_{-\infty}^{t} m(\tau)d\tau\right]$$

where

$$\left|2\pi k_f \int_{-\infty}^{t} m(\tau)d\tau\right| \ll 1$$

- We say that $s_{FM}(t)$ is a narrowband FM signal.
- For example, consider when $m(t) = A_m \cos(2\pi f_m t)$.

$$s_{FM}(t) = A_c \cos\left(2\pi f_c t + \frac{A_m k_f}{f_m} \sin(2\pi f_m t)\right)$$

$$s_{FM}(t) = A_c \cos\left(2\pi f_c t + \beta_F \sin(2\pi f_m t)\right)$$

Narrowband FM Approximation

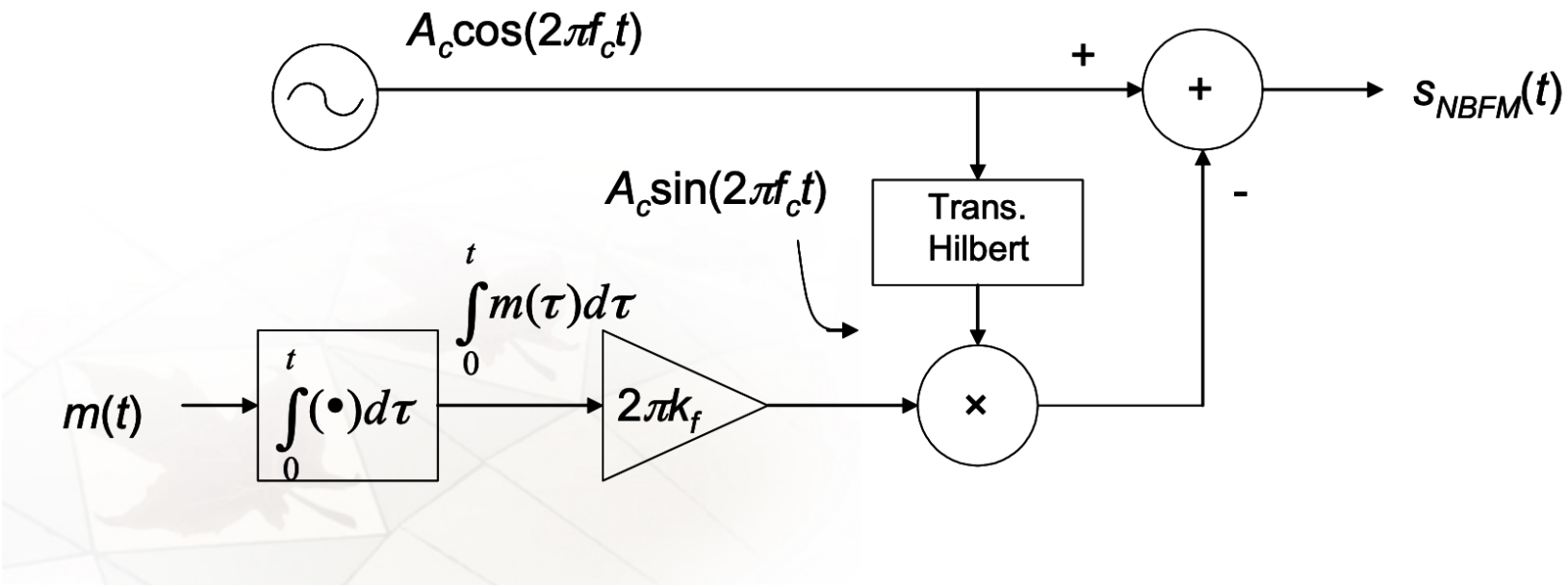
- When $\beta_F \ll 1$, the FM signal is NBFM.
- Using $\cos(A+B) = \cos A \cos B - \sin A \sin B$, we write

$$s_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int_0^t m(\tau)d\tau\right]$$

$$= A_c \cos(2\pi f_c t)\cos\left(2\pi k_f \int_0^t m(\tau)d\tau\right) - A_c \sin(2\pi f_c t)\sin\left(2\pi k_f \int_0^t m(\tau)d\tau\right)$$

$$\approx A_c \cos(2\pi f_c t) - A_c\left(2\pi k_f \int_0^t m(\tau)d\tau\right)\sin(2\pi f_c t)$$

(since for small A, $\cos A \approx 1$, $\sin A \approx A$)


$A_c\cos(2\pi f_c t)$
+
+
$s_{NBFM}(t)$
$A_c\sin(2\pi f_c t)$
Trans.
Hilbert
-
$\int_0^t m(\tau)d\tau$
$m(t)$
$\int_0^t (\bullet)d\tau$
$2\pi k_f$
×